

BPM501
ER Module
IMCQ 4 - KEY

Spring 2026

A 12-year-old girl is brought to the physician because of a 6-month history of facial hair development. Her mother explains that for the past 6 months she has experienced changes to her genital area. Physical examination shows no signs of breast development. Pelvic examination shows an enlarged clitoris and signs of partial fusion of the labia majora and absent vaginal opening. Bilateral masses are palpable in the inguinal canal. Ultrasonography shows absence of the uterus, ovaries and vagina. Laboratory studies show levels of testosterone consistent with a peripubertal boy. Chromosomal analysis confirms the presence of a Y chromosome. Which of the following best describes the development of cells that produce the enzyme required for fetal masculinization of the external genitalia?

- A. Medullary cells of the indifferent gonad
- B. Primordial germ cells
- C. Primary sex cords
- D. Mesenchyme surrounding seminiferous cords
- E. The cells lining the seminiferous cords

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A 10-day-old female newborn is brought to the physician by her mother because of constant irritability and redness and swelling in her genital region. Physical examination shows fecal matter in the vaginal canal. Further investigation shows the presence of a rectovaginal fistula. Which of the following developmental processes most likely failed to develop adequately, resulting in her condition?

- A. The urogenital sinus is separated from the anorectal canal by the paramesonephric ducts
- B. The urorectal septum divides the cloaca into the urogenital sinus and the anorectal canal
- C. The mesonephric ducts fuse to form the urogenital sinus and anorectal canal
- D. The cloaca remains unchanged throughout life

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A 25-year-old woman comes to the emergency department because of lower abdominal and pelvic cramps for the past 4 hours. She is 14 weeks pregnant with her first child. **Ultrasonography** shows blood in the uterus and a division of the uterine cavity. She is diagnosed with a spontaneous miscarriage due to a septate uterus. Which of the following best describes the mechanism by which this developmental abnormality forms?

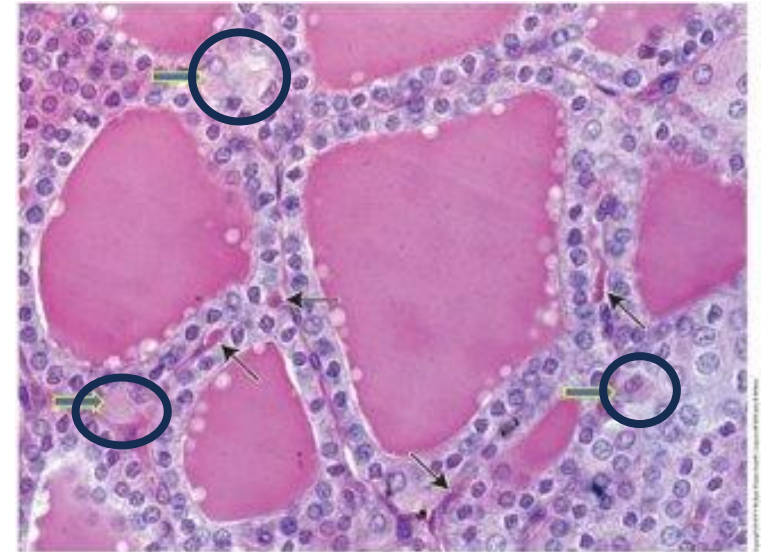
- A. Failure of complete fusion of the paramesonephric ducts
- B. Failure of fusion of the mesonephric ducts
- C. Failure of uterine septum to regress
- D. Incomplete development of one of the mesonephric ducts
- E. One paramesonephric duct is inhibited and this duct does not fuse with the second duct

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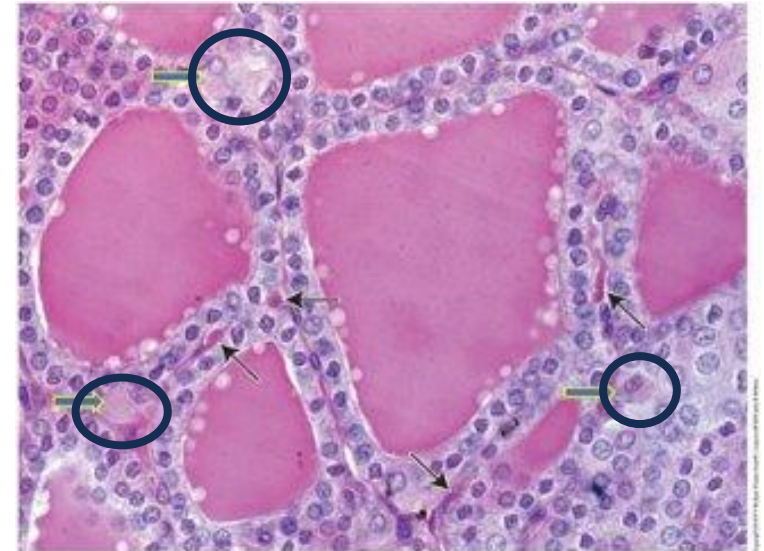
A 62-year-old woman comes to the physician because of a 3-month history of mild muscle cramps in her legs and tingling in her fingers. Laboratory studies show mild hypocalcemia despite adequate dietary calcium intake, and elevated calcitonin levels. Ultrasonography shows a mass on her thyroid gland. Examination of a biopsy specimen of the mass shows a malignancy arising from the circumscribed cells in the photomicrograph shown. Which of the following will also most likely be observed in this patient?

- A. Suppressed levels of thyrotropin-releasing hormone (TRH)
- B. Increased conversion of thyroglobulin into active thyroid hormones
- C. Decreased levels of thyroid stimulating hormone (TSH)
- D. Increased synthesis and secretion of parathyroid hormone (PTH)
- E. Hyperplasia of the follicular cells



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A 49-year-old man comes to the physician because of a 3-year history of infertility. Physical examination shows no abnormalities. A biopsy specimen of his testes show distorted morphology of a columnar shaped cell with a pale nucleus and prominent nucleolus, that spans the full thickness of the seminiferous epithelium. Given the affected cell type, which of the following functions will most likely be affected?

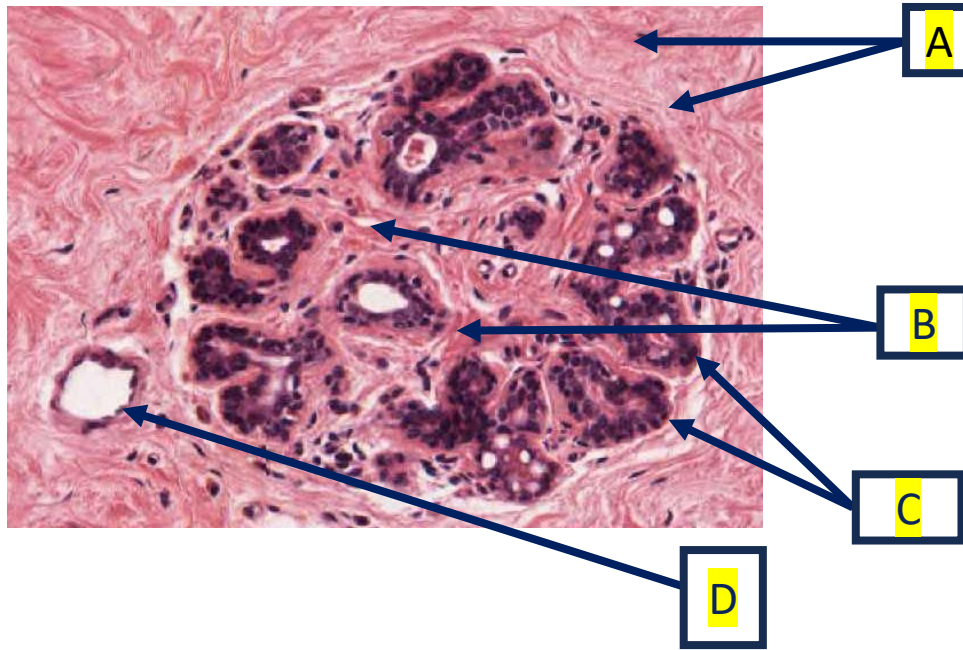
- A. Synthesis of FSH
- B. Production of testosterone
- C. Secretion of inhibin
- D. Direct transport of spermatozoa
- E. Production of semen

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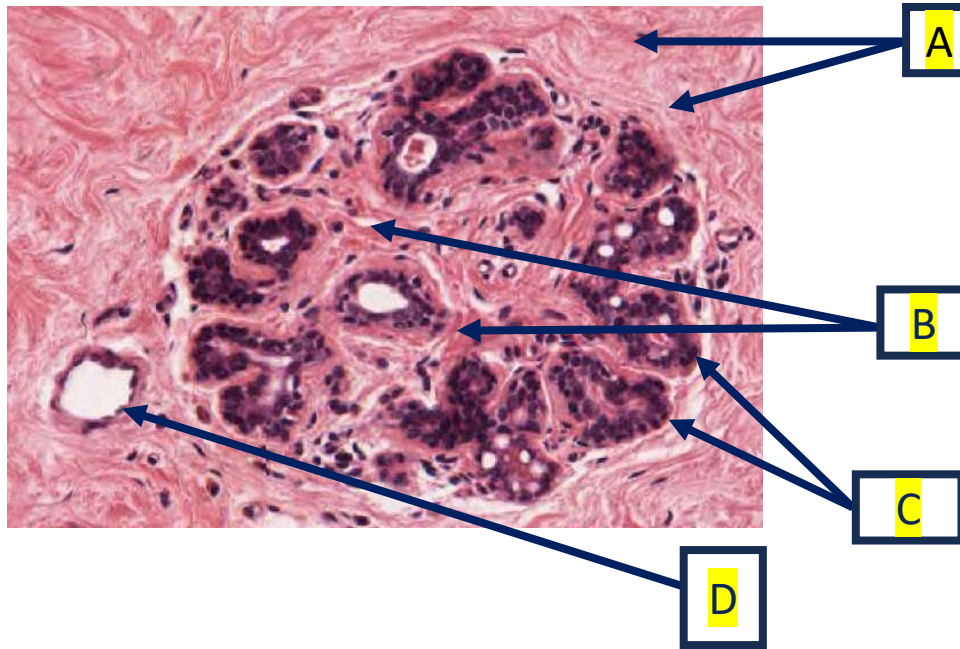
A 22-year-old nulliparous woman comes to the physician because of a 4-month history of a lump in her right breast. She is on day 22 of a 28-day menstrual cycle. Physical examination shows a well-defined 3-cm mass in the upper lateral quadrant of the right breast. Microscopic examination of a biopsy specimen of the mass shows a carcinoma originating from the terminal ductules. Which of the following labels in the attached micrograph of a normal breast tissue best identifies the structure from which this patient's cancer originated?

- A. A
- B. B
- C. C
- D. D



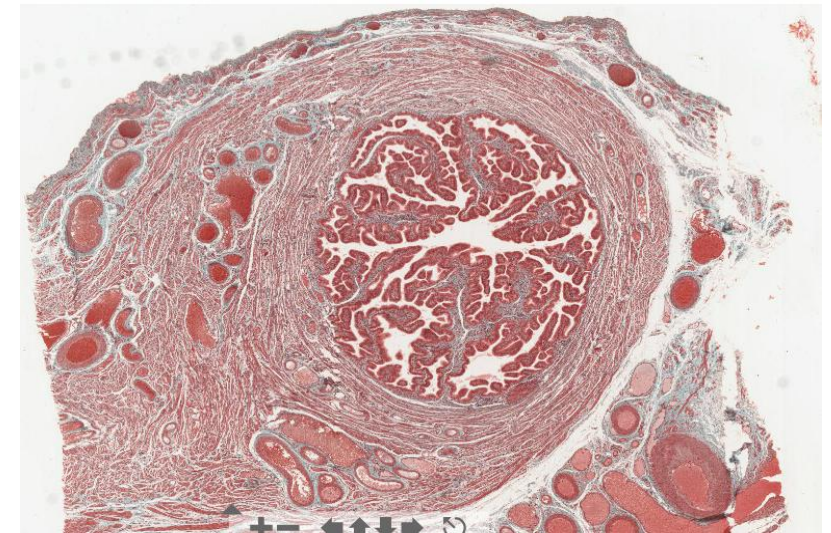
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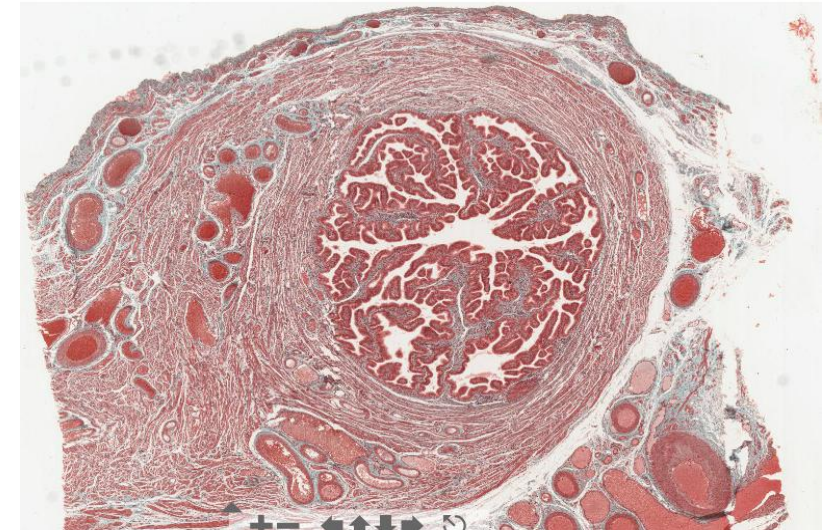
A 23-year-old woman comes to the physician because of a 6-month history of recurrent pelvic pain, experienced primarily around her menstrual periods and during sexual intercourse. She denies any sexually transmitted infections or recent pelvic surgery. Physical examination shows a fixed, anteverted uterus and nodularity in the posterior cul-de-sac (Pouch of Douglas). Examination of a biopsy specimen of the nodules show endometrial glands and stroma outside the uterus with implants involving the structure shown in the attached photomicrograph. Which of the following statements best describes this structure?

- A. Mucosa is often lost during the menstrual cycle
- B. Readily invaded by syncytiotrophoblastic cells during implantation
- C. Epithelium consists of Peg and ciliated columnar cells
- D. Epithelial cells secrete the hormone hCG
- E. Degenerates to form the corpus luteum after expulsion of the secondary oocyte



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A 2-week-old female newborn is referred to a pediatric immunologist after having a positive newborn screening result for severe combined immunodeficiency (SCID). She was born at term with no complications. Family history is negative for immunodeficiency. Confirmatory tests show less than 1% adenosine deaminase activity. She is currently asymptomatic and has not received any vaccinations. Which of the following is the best current method to cure this patient?

- A. Administration of vaccines
- B. Begin prophylactic antibiotics
- C. Enzyme replacement therapy
- D. Bone marrow transplant
- E. Gene therapy

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SOM.MK.III.BPM2.4.ER.3.GNET.1019; Appreciate the need to provide appropriate assistance in the decision-making process of patients when faced with a positive result and further testing

- A. Administration of vaccines
- B. Begin prophylactic antibiotics
- C. Enzyme replacement therapy
- ✓ D. Bone marrow transplant
- E. Gene therapy

A male infant is born via emergency cesarean section for a prolapsed cord at 33 weeks' gestation to a 22-year-old primigravida. The mother had a 3-week history of severe nausea and vomiting, and a diagnosis of acute fatty liver of pregnancy (AFLP) was made. The newborn weighs 1,630 g at birth, which is appropriate for his gestational age. In the neonatal intensive care unit, he shows continued bleeding from the venipuncture sites. Coagulation studies show that the prothrombin time (PT) is 98.5 seconds (12-14.7 seconds), and the aPTT is 104 seconds (23-38 seconds). Which of the following best explains the findings in this patient?

- A. Vitamin K deficiency
- B. Folate deficiency
- C. Iron deficiency
- D. Lactose intolerance
- E. Essential amino acid deficiency
- F. Vitamin D deficiency

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SOM.MK.III.BPM2.1.ER.2.BCHM.1058 Identify the age range of neonate and infant.
Discuss Vitamin K deficiency bleeding (VKDB) in the neonate and infant.

A 4-month-old girl is brought to the physician by her mother because of a 2-day history of drowsiness and poor feeding. Her mother says she often needs to be awakened to nurse. She was born spontaneously at term with body weight 3500 grams. She is drowsy with a hoarse cry. Physical examination shows scanty, coarse and brittle hair, a puffy face with a thick broad tongue and hypotonia. Which of the following maternal dietary micronutrient deficiencies during pregnancy is the most likely risk factor for the development of the patient's condition?

A. Zinc

B. Folate

C. Iron

D. Calcium

E. Iodine

F. Vitamin A

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SOM.MK.III.BPM2.1.ER.2.BCHM.1063; Review the need for nutritional balance and the consequences of deficiencies on mother, fetus and infant as it relates to: Energy requirements and weight gain (Including carbohydrates and Omega 3 fatty acids like DHA), Proteins (Including enzymes and immunoglobulins in breast milk), Iron (Microcytic iron deficiency anemia), Folate (Macrocytic anemia and neural tube defects), Calcium, Vitamin A, Zinc and Iodine (congenital hypothyroidism).

A 5-month-old boy is brought to the physician by his parents because of a 1-month history of poor weight gain. His weight is 4.5 kg (<- 3 SD), his length is 57 cm (<- 3 SD), and both his weight-for-height and height-for-age are decreased. Physical examination shows an irritable child with shrunken arms, thighs, and buttocks. Skin examination shows redundant skin folds and thin, sparse, light-colored hair that's easily plucked off. Which of the following sets of metabolic changes are most likely expected in this patient?

		Cortisol	Insulin	Gluconeogenesis
A.	I	↓	↓	↓
B.	II	↓	↓	↑
C.	III	↓	↑	↑
D.	IV	↑	↑	↑
E.	V	↑	↓	↑

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D.	IV	↑	↑	↑
 E.	V	↑	↓	↑

A 51-year-old woman comes to the physician because of an 18-month history of irregular menses, difficulty concentrating at work, and hot flashes. Her last menstrual period was 6 months ago. Vital signs are within normal limits. Genital examination shows atrophy of the labia, loss of vaginal rugae, and decreased vulvovaginal secretions. She is diagnosed with menopause. Deficiency of which of the following hormones best explains this patient's presentation?

- A. Estrogen
- B. Follicular stimulating hormone
- C. Luteinizing hormone
- D. Progesterone
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SOM.MK.II.BPM2.1.ER.1.PHYS.1081. Explain the age-related changes in the hypothalamo-pituitary-ovarian axis in females that lead to puberty, reproductive maturity, and reproductive senescence (menopause).

An 18-year-old woman is brought to the physician by her mother because of failure to begin menses. The mother says her classmates had menarche at the age of 13 years. She has been a swim champion in the school for the last 5 years and does vigorous practice daily. She is 175 cm (5 ft 9 in) tall and weighs 57 kg (125 lb); BMI is 18.5 kg/m² (Normal: 19-25). Vital signs are within normal limits. Physical examination shows scanty pubic and axillary hairs, underdeveloped external genitalia, and underdeveloped breasts. Laboratory studies show low concentrations of gonadotropins and ovarian hormone concentrations. Other pituitary hormones are within reference ranges. She is diagnosed with functional hypothalamic amenorrhea. Which of the following terms best explains this patient's condition?

- A. Primary hypogonadism
- B. Secondary hypogonadism
- C. Primary hypergonadism
- D. Secondary hypergonadism
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SOM.MK.II.BPM2.1.ER.1.PHYS.1083. Explain age related changes in hypogonadism in females.

A 28-year-old woman comes to the physician for a routine examination. She is married recently and couple decided to have children. Her menses occur at regular 34-day intervals and last for 4-5 days. She has no chronic illness. Vital signs are within normal limits. Physical examination shows normal external genitalia, normal uterus, and normal adnexa. Serum hormone studies are within the reference ranges. The physician teaches her to calculate the day of ovulation to achieve the best possibility of conception. Which of the following days is most likely the time of ovulation in this patient?

- A. 5th day of cycle
- B. 10th day of cycle
- C. 15th day of cycle
- D. 20th day of cycle
- E. 26th day of cycle

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SOM.MK.II.BPM2.1.ER.1.PHYS.1077.Describe the hormonal changes, follicular development, ovulation formation & atresia of corpus luteum throughout the ovarian cycle

A 38-year-old woman at 32 weeks' gestation comes to the office because of a 2-week history of swelling in her legs. She says the swelling appears when she sits for a long time and disappears by elevating legs or lying supine. She has no history of shortness of breath or passing frothy urine. Her blood pressure is 128/64 mm Hg; other vital signs are within normal limits. Physical examination shows no abnormalities and lungs are clear to auscultation. Urinalysis shows no protein. The physician assures that the swelling in the leg-down position is normal during term because of the gravid uterus compressing venous drainage from the legs. Her pre-pregnancy medical records show a blood pressure of 120/80 mm Hg. Which of the following mechanisms best explains the reduced diastolic blood pressure in this patient?

- A. Decrease in peripheral venous distention
- B. Decrease in peripheral vascular resistance
- C. Increase in preload
- D. Increase in blood volume
- E. Decrease in stroke volume

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SOM.MK.II.BPM2.1.ER.1.PHYS.1087. Explain the maternal and fetal endocrinology and physiological changes during pregnancy.

A 24-year-old woman at 28 weeks' gestation comes to the office for a prenatal visit. Vital signs are within normal limits. Fundal height and fetal heart tones are also normal. Laboratory studies show an elevated serum glucose concentration. The result of an oral glucose tolerance test shows an elevated serum glucose concentration even 2 hours after a glucose load. She is diagnosed with gestational diabetes. Which of the following is the most likely physiologic effect of the hormone that is responsible for this patient's condition?

- A. Increases maternal insulin sensitivity
- B. Increases maternal fat mobilization to fetus
- C. Decreases placental storage of fuel
- D. Inhibits maternal milk secretion
- E. Stimulates maternal prolactin secretion

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SOM.MK.II.BPM2.1.ER.1.PHYS.1089. Explain how gestational diabetes can lead to problems during pregnancy.